

Appendix

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A RL Teacher Training and Safety Details

Table 1 contrasts the actor’s noisy, deployable signals with the centralized critic’s clean, privileged simulator state. The palm and five fingertips constitute the six hand links. A “twist” is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	87	133	

“–” denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

Termination conditions. All conditions are checked at 60 Hz in the environment frame; the table-top height is $z_T = 0.25$ m.

- Harmful collision:* We define a safety-clearance region above the table and monitor all arm–table contacts. We trigger this termination when the palm or fingertips enter the clearance region, or when any arm link contacts the table.
- Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.
- Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90° , at which point the nominal grasp geometry is inverted.
- Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.
- Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we use the labeling priority object escape $\succ$ hand too far $\succ$ harmful collision $\succ$ palm flipped $\succ$ timeout.

Runs and random seeds. Teacher performance in Main Paper Table II is evaluated over 200 episodes per object, and the real-world results are evaluated over 30 episodes per object. Each of the 40 object-specific teacher policies is trained with five independent random seeds $\{1, 2, 3, 4, 5\}$, with the same evaluation protocol applied to every seed. For each seed, teacher performance is evaluated on all 40 objects. The table below reports the mean $\pm$ standard deviation across the 40 object-level rates for lift success, overall unsafe events, and the four unsafe-event categories separately for each seed.

Table 2: Teacher policy performance across all 40 objects for each random seed (200 evaluation episodes per object and seed). Each entry is the mean $\pm$ standard deviation across the 40 object-level rates. Numerical entries are provisional synthetic placeholders pending verification with the measured evaluation data.

Metric (%)	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Lift success	86.7 ± 7.8	87.4 ± 7.5	86.9 ± 8.0	87.8 ± 7.3	87.1 ± 7.7
Unsafe rate avg.	14.2 ± 6.1	13.6 ± 5.8	14.0 ± 6.3	13.2 ± 5.6	13.8 ± 6.0
Harmful collision	3.1 ± 3.5	2.9 ± 3.3	3.2 ± 3.6	2.8 ± 3.2	3.0 ± 3.4
Object escape	9.0 ± 5.2	8.8 ± 5.0	8.9 ± 5.3	8.7 ± 4.9	8.9 ± 5.1
Palm flipped	1.2 ± 2.4	1.1 ± 2.3	1.1 ± 2.3	1.0 ± 2.2	1.1 ± 2.3
Hand too far	0.9 ± 1.9	0.8 ± 1.8	0.8 ± 1.8	0.7 ± 1.7	0.8 ± 1.8

Teacher safety and disagreement handling. Main Paper Table II reports representative teacher unsafe-event rates. Across all 40 teacher policies, unsafe-event rates range from 5–30%. Because an intervention transfers control to an imperfect teacher, the framework’s safety performance is restricted by the teacher’s own performance ceiling. Teacher takeover can reduce unsafe exploration, but it does not provide a formal safety guarantee. The controller does not screen the teacher and student as two symmetric candidates. Instead, the Bellman critic evaluates only the student state–action pair, and the deterministic intervention gate is

$$g_t = \mathbb{I}[\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q], \quad a_t = (1 - g_t)a_t^S + g_t a_t^T.$$

Consequently, disagreement between the two intervention signals is resolved conservatively: either excessive teacher–student action disagreement or a low student success-value prediction is sufficient to trigger teacher takeover. The student action is executed only when both tests accept it. Once triggered, teacher control is retained for a preset minimum dwell interval to avoid rapid switching between policies.

Because the critic is not used to veto the teacher action, the implemented gate has no branch in which both candidate actions are rejected. The teacher action is the terminal action-level fallback whenever $g_t = 1$, even though the teacher is empirically rather than certifiably safe. If the resulting rollout reaches any monitored unsafe condition, the simulator terminates and resets the episode according to the termination rules above; the controller does not synthesize a third recovery action. Non-finite policy outputs are treated as a runtime error before the environment step rather than being accepted by the gate. This design makes the teacher’s performance ceiling an explicit limitation of the intervention mechanism.

B VLM Prompt and Inference Details

Visual-input storage and example. Candidate visual inputs are retained during a run in a bounded in-memory buffer of at most 64 samples. At each query, up to eight selected samples are embedded as JPEG data URLs in a self-contained `<run_dir>/vlm_reports/*.html` report together with their metadata, the prompt, model reply, parsed JSON, and API diagnostics. The accompanying `vlm_threshold_advisor.jsonl` strips the base64 payloads and retains only metadata; no separate image files are written.

Figure 1 shows four actual inputs extracted from the first archived query report, one for each object in that rollout.

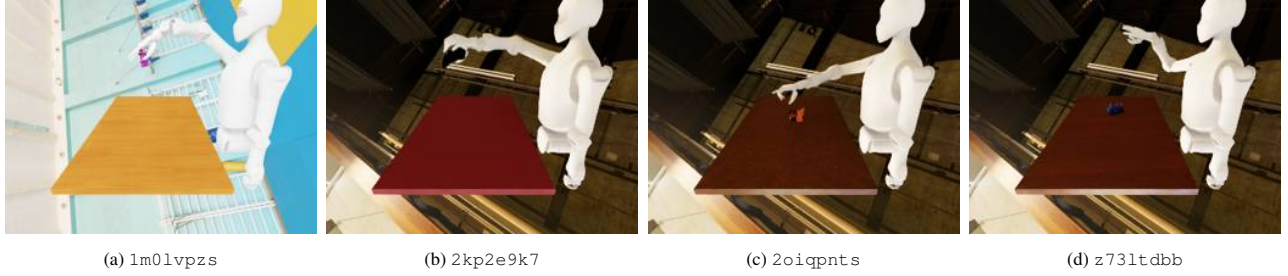


Figure 1: Representative left-camera RGB inputs embedded in `step_000002000_attempt_0001_success.html`. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`. The transmitted images are 256×192 JPEGs (quality 70). This archived call tagged the images as `detail="low"`; new calls transmit the same image format with `detail="high"`.

The following is the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

System message: You tune safety thresholds and return compact JSON only. User message: You are a conservative threshold advisor for DexSafeDagger robot distillation. Do not decide intervention directly. Recommend thresholds for the existing gates. Task: Tune teacher-student disagreement and predicted-success thresholds for dexterous robot distillation. Existing arbitration: - intervene if <code>teacher_student_l2</code> $\geq$ <code>l2_threshold</code> - or if <code>predictor_success</code> $\leq$ <code>success_threshold</code> Return only strict JSON with keys: <code>l2_threshold</code>, <code>success_threshold</code>, <code>confidence</code>, <code>reason</code>. Keep changes conservative. Prefer small threshold shifts unless the recent window shows clear over- or under-intervention. Base thresholds and clamps: - <code>base_l2_threshold</code>={<code>base_l2_threshold</code>} - <code>allowed_l2_range</code>=[{<code>l2_min</code>}, {<code>l2_max</code>}] - <code>base_success_threshold</code>={<code>base_success_threshold</code>} - <code>allowed_success_range</code>=[{<code>success_min</code>}, {<code>success_max</code>}] Current advisor thresholds: - <code>current_l2_threshold</code>={<code>current_l2_threshold</code>} - <code>current_success_threshold</code>={<code>current_success_threshold</code>}	Recent rollout statistics: <pre>{ "frame": {frame}, "intervention_rate": {beta}, "step": {step}, "sample_count": {N}, "l2_threshold": {tau_delta}, "success_threshold": {tau_Q}, "unsafe_episode_rate": {r_unsafe}, "l2": {"min": {v}, "max": {v}, "mean": {v}, "p10": {v}, "p50": {v}, "p90": {v}}, "success": {"min": {v}, "max": {v}, "mean": {v}, "p10": {v}, "p50": {v}, "p90": {v}}, "unsafe_reason_prop": { "hand_too_far": {v}, "harmful_collision": {v}, "object_out_of_bound": {v}, "palm_flipped": {v}}, "visual_buffer_size": {B}, "visual_samples_attached": {M}, "visual_samples": [{ "env_id": {v}, "frame": {v}, "image_attached": true, "image_key": {v}, "source": {high_l2 low_success unsafe_triggered}, "l2_threshold": {v}, "success_threshold": {v}, "teacher_student_l2": {v}, "predictor_success": {v}, "object_id": {v}, "object_name": {v}, "object_env_count": {v}, "step": {v}, "unsafe": {true false}}, ...] }</pre> For each selected visual sample $i=1, \dots, M$: Visual sample {i} metadata: {the corresponding record above, excluding "image_attached"} <IMAGE {i}: JPEG data URL, detail="high">
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The earlier abbreviated prompt omitted the distribution summaries, threshold clamps, unsafe-reason breakdown, visual-buffer context, and per-image metadata; it also used the nonexistent output key `disagreement_threshold` and omitted confidence. The runtime prompt uses `l2_threshold`. For reproducibility, the template above preserves the literal inclusive comparison symbols sent to the VLM. The executed controller uses the strict tests in the intervention equation above; the distinction affects only exact equality at a threshold and does not change the VLM’s role as a threshold advisor.

Model and input configuration. We used the OpenAI GPT-5.5 multimodal model through the Chat Completions endpoint (`/v1/chat/completions`, `model="gpt-5.5"`). Each request set `stream=false` and `max_completion_tokens=1024`; no temperature or reasoning-effort parameter was sent, so the provider defaults applied. The model received the textual rollout statistics above and up to eight RGB images, encoded as downsampled JPEG data URLs with high visual detail. We issued the first VLM query after 2,000 online distillation steps and subsequently queried it every 10,000 steps, resulting in 10 calls over the 100,000-step run. The archived API reports,

which were generated with the previous low-detail image setting, record 30,641 prompt tokens and 4,841 completion tokens, or 35,482 tokens in total.

Output processing and threshold updates. The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "l2_threshold": <number>, "success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The implementation first attempts to parse the complete assistant response as JSON. If that fails, it parses the substring from the first opening brace to the last closing brace, which permits recovery from a surrounding Markdown code fence or brief prefatory text. Only a JSON object is accepted. The two threshold fields are converted to floating-point values independently; a missing or non-convertible field leaves that threshold at its current value, and unrecognized fields are ignored. The `confidence` and `reason` fields are recorded for diagnostics but neither changes nor vetoes the controller update.

The returned thresholds are interpreted as absolute candidates, not direct intervention decisions. For each $x \in \{\delta, Q\}$, the candidate is clipped and exponentially smoothed as

$$\bar{\tau}_x^{(k)} = \text{clip}\left(\hat{\tau}_x^{(k)}, c_x^{\min} \tau_x^{(0)}, c_x^{\max} \tau_x^{(0)}\right), \quad \tau_x^{(k+1)} = (1 - \alpha) \tau_x^{(k)} + \alpha \bar{\tau}_x^{(k)}.$$

We used $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. In the archived run, $\tau_\delta^{(0)} = 3.0$, yielding the allowed range $[1.5, 4.5]$. Warm-start calibration set $\tau_Q^{(0)} = 0.7081624$, yielding the implemented range $[0.3540812, 1.0622436]$. For example, the first recommendation $\hat{\tau}_\delta = 3.2$ produced the applied value $0.9(3.0) + 0.1(3.2) = 3.02$.

In `active` mode, the filtered disagreement threshold becomes the controller’s τ_δ , and the filtered success threshold is assigned to the Bellman critic’s intervention threshold. They affect the next control step through the strict gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q therefore increases teacher intervention. In `shadow` mode, the same outputs are logged without changing either threshold. The first query was made at online distillation step 2,000 and later queries every 10,000 steps, at steps 2,000, 12,000, ..., 92,000.

The deterministic fallback is to retain the last applied thresholds. This occurs when the API key is unavailable, the request times out or fails, the API response is malformed, or no JSON object can be parsed. If only one threshold field is missing or non-convertible, that coordinate is retained while a valid coordinate may still be clipped and updated. A failed attempt is logged and is not retried immediately; the next attempt occurs at the next scheduled update. Finally, the VLM and this adaptation mechanism are used only during simulation-based online distillation. There is no real-world student training, and during physical deployment the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation.

C Additional Experimental Results

Bellman critic evaluation. [State how safety risk and task success are separated, or justify the retained target. Report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate for unsafe episodes at the deployed threshold.]

Training and resource comparison. All ablations use the same training and computational resource budget and differ only in the intervention mechanism.

Twenty-two-object inventory. The complete evaluation set comprises the following 22 objects (the six objects reported in the main paper are marked with an asterisk):

1	WindCart*	12	Ogre
2	Boot*	13	Cheburashka
3	ToolBox*	14	Lucy statue
4	Cow*	15	Beethoven bust
5	Plane*	16	Igea bust
6	Shoe*	17	Dog pair
7	Horse	18	Spot dog
8	Teapot	19	Monkey head
9	Sphinx	20	Blub character
10	Dragon	21	Statue
11	Homer bust	22	Dinosaur

Complete real-world evaluation. We will report results for the remaining 16 real-world test objects that are not shown in the manuscript’s six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes. The entries below are provisional synthetic values for layout checking and will be replaced with the measured results.

Table 3: Provisional per-object results for the remaining 16 objects (30 episodes per object). Synthetic values are shown for layout checking only; **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Cheburashka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	13.3	23.3	20.0	26.7	16.7	10.0	23.3
Harmful collision	3.3	0.0	10.0	6.7	13.3	3.3	3.3	6.7
Object escape	13.3	13.3	13.3	13.3	10.0	13.3	6.7	13.3
Palm flipped	0.0	0.0	0.0	0.0	3.3	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	Dog Pair	Spot	Monkey head	Blub	Statue	Dinosaur
Lift Success	86.7	73.3	70.0	63.3	73.3	60.0	66.7	56.7
Unsafe Rate Avg	20.0	30.0	33.3	40.0	30.0	43.3	36.7	46.7
Harmful collision	6.7	13.3	16.7	20.0	13.3	23.3	16.7	26.7
Object escape	13.3	13.3	13.3	16.7	13.3	16.7	16.7	16.7
Palm flipped	0.0	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

[Replace all provisional synthetic values with the measured results for the remaining 16 objects. Report 95% confidence intervals and statistical comparisons, and provide aggregate results separately for seen and unseen objects across the full 22-object evaluation.]

Physical baseline comparison. [Report the feasible matched real-robot baseline using the same objects, initial-condition protocol, and number of trials, including lift success, unsafe-event incidence, uncertainty, and the statistical comparison to VLM-DexSafeDagger.]